

ONLINE WORKSHOP

New Horizons of Drug Discovery

Molecular Docking

6 - 8 September 2026 | 7:00 PM – 9:00 PM
Organised by Fluxiora Learning | Fluxiora Intelligence Private Limited

Workshop Overview

This intensive three-day online workshop is designed to equip participants with both the theoretical foundations and practical skills required for modern computational drug discovery. Through a structured curriculum spanning molecular docking fundamentals to cutting-edge machine learning applications, attendees will gain hands-on experience with industry-standard tools and software platforms.

The workshop bridges the gap between classical bioinformatics and contemporary AI-driven approaches, making it ideal for graduate students, researchers, and professionals seeking to advance their computational drug discovery capabilities.

Event Details

Event Title	Online Workshop on New Horizons of Drug Discovery: Molecular Docking
Format	Online (Virtual)
Dates	6 – 8 September 2026
Duration	3 Days
Timings	7:00 PM – 9:00 PM (2 hours per day)
Total Hours	6 hours of instruction
Organiser	Fluxiora Learning
Registered Body	Fluxiora Intelligence Private Limited
Instructor	Mr. Utkarsh Tripathi (PhD Scholar, IIT Kanpur)
Fees	399/-

About the Instructor



Mr. Utkarsh Tripathi

PhD Scholar | Indian Institute of Technology Kanpur (IIT Kanpur)

Mr. Utkarsh Tripathi is a PhD scholar at the prestigious Indian Institute of Technology Kanpur (IIT Kanpur), one of India's foremost institutes of science and technology. His research expertise spans computational chemistry, structural bioinformatics, and the application of machine learning methods to drug discovery pipelines.

With deep experience in molecular docking methodologies — including blind docking, targeted docking, and AI-driven binding site prediction — Mr. Tripathi brings both rigorous academic knowledge and practical computational expertise to this workshop.

Workshop Curriculum

Day 1	Introduction to Molecular Docking & Protein-Inhibitor Interactions
1	Fundamentals of molecular docking — principles and scope
2	Structural biology concepts: proteins, ligands, and binding sites
3	Protein–inhibitor interactions: types and significance
4	Scoring functions and energy minimisation
5	Overview of computational drug discovery workflow
6	Introduction to key software environments

Day 2	Blind Docking with AutoDock, ChimeraX, PyRX & DiffDock
1	Concept of blind docking — whole-protein search space
2	Hands-on: Protein preparation using UCSF ChimeraX
3	Ligand preparation and grid box configuration
4	Running blind docking simulations with AutoDock / PyRX

5	AI-powered docking with DiffDock — diffusion-based approach
6	Results analysis, visualisation, and interpretation

Day 3	Targeted Docking, ML with PrankWeb & Future Directions
1	Targeted docking: defining and refining binding pockets
2	Binding site prediction with PrankWeb (machine learning approach)
3	Comparative analysis: blind vs. targeted docking outcomes
4	Introduction to Molecular Dynamics (MD) Simulations
5	Future directions: generative AI in drug discovery
6	Q&A, discussion, and career pathways in computational drug discovery

Learning Outcomes

Upon successful completion of this workshop, participants will be able to:

- Understand the theoretical principles underpinning molecular docking and protein–ligand interactions
- Prepare protein and ligand structures for docking using UCSF ChimeraX and PyRX
- Perform and interpret blind docking experiments using AutoDock and DiffDock
- Predict binding sites computationally using the ML-powered PrankWeb platform
- Conduct targeted docking with a well-defined binding pocket
- Appreciate the role of Molecular Dynamics (MD) Simulations in validating docking results
- Navigate the emerging landscape of AI and generative models in drug discovery

Software & Tools Covered

Tool / Software	Purpose
AutoDock	Industry-standard molecular docking engine
UCSF ChimeraX	Molecular visualisation and protein preparation
PyRX	User-friendly docking interface for AutoDock/Vina
DiffDock	AI-powered diffusion model for blind docking
PrankWeb	Machine learning-based binding site prediction
MD Simulations	Post-docking validation and dynamics analysis (overview)

Who Should Attend

- Undergraduate & postgraduate students
 - PhD scholars & researchers
 - Students from **Life Sciences, Biotechnology, Biochemistry, Pharmacy, Chemistry, Chemical Engineering, Bioinformatics & related fields**
 - Academicians & industry professionals
 - Basic knowledge of Biology/Chemistry recommended
 - **No prior molecular docking experience required**
 - Anyone with a foundational interest in learning molecular modelling and docking techniques
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About the Organiser

Fluxiora Learning

A division of Fluxiora Intelligence Private Limited

Fluxiora Learning is the educational arm of Fluxiora Intelligence Private Limited, dedicated to delivering high-quality, research-oriented training programmes in emerging scientific and technological domains. Through expert-led workshops, bootcamps, and online courses, Fluxiora Learning empowers the next generation of scientists and innovators with practical, applicable skills.

Contact Us

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This document is an official workshop brief prepared by Fluxiora Intelligence Private Limited. For registrations or enquiries, please contact the organiser.

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